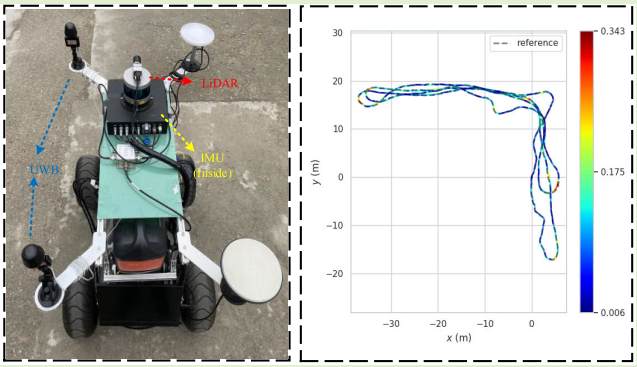
Uncertainty-Aware UWB/LiDAR/INS Tightly Coupled Fusion Pose Estimation via Filtering Approach

Hong Liu, *Graduate Student Member, IEEE*, Shuguo Pan, Pengbo Wu, Kegen Yu, *Senior Member, IEEE*, Wang Gao, and Baoguo Yu

***Abstract*—Precise and reliable pose estimation is a critical requirement for autonomous system. In recent years, light detection and range (LiDAR)-inertial odometry (LIO) has made significant advancements, especially in challenging environments with varying illumination and other complexities. However, LIO systems is known to local navigation, there can be a problem of error accumulation over time, particularly in global navigation satellite system (GNSS)-denied environments or in the absence of prior maps. Therefore, integrating the ultra-wideband (UWB) technique can effectively correct long-term state drift and enhance system performance, making it a promising solution for the demanding estimation task. In this contribution, we propose the tightly coupled method to fuse point cloud, inertial measurement, and UWB ranging information via iterative error state Kalman filtering (IESKF). With the UWB-aid initialization, the global-type and drift-free initial state can be obtained, which also facilitates the convergence of solution. We establish uncertainty-aware models and derive observation covariance for sensors, which play a crucial role in the heterogeneous multisensor fusion for pose estimation. Furthermore, extensive experiments in various scenarios are conducted using a customized platform. The results demonstrate that our approach can provide robust and consistent trajectory and mapping results while keeping computational costs low, even in the case when laser degeneration, non-line-of-sight (NLOS) ranging or limited UWB anchor stations.**



***Index Terms*—Filtering method, pose estimation, ultra-wideband (UWB)/light detection and range (LiDAR)/inertial navigation system (INS) integration, uncertainty-aware.**

NOMENCLATURE

Notations	*Meaning*
$P^W_{\text{anchor},\,i}$	Anchor node i position in the world frame.
$\hat{x}, \bar{x}, x$	Predicted, updated, and true value of x.
R^A_B, q^A_B	Rotation from B frame to A frame.
t^A_B	Translation from B frame to A frame.
b_w, b_a	Bias of IMU gyroscope and accelerometer.
n_w, n_a	Noise of IMU gyroscope and accelerometer.
$(\cdot)^T$	Transpose operation.
$(\cdot)^{-1}$	Inverse operation.
$\boxplus/\boxminus$	Boxplus and boxminus operator in manifold.

Manuscript received 16 December 2023; accepted 2 February 2024. Date of publication 14 February 2024; date of current version 2 April 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3900804, in part by the Research Fund of Ministry of Education of China and China Mobile under Grant MCM20200J01, and in part by the Fundamental Research Funds for the Central Universities under Grant 2242021R41134. The associate editor coordinating the review of this article and approving it for publication was Prof. Meribout Mahmoud. *(Corresponding author: Shuguo Pan.)*

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Digital Object Identifier 10.1109/JSEN.2024.3362741

I. INTRODUCTION

RELIABLE and accurate pose estimation is the fundamental capability for autonomous unmanned systems such as autonomous driving [1], warehouse logistics [2], and home service robots [3]. From previous studies, real-time kinematic (RTK) and precise point positioning (PPP) are two strategies from the global navigation satellite system (GNSS), which can provide centimeter-level positioning accuracy in outdoor open scenarios [4]. However, the performance intricately hinges on signal quality and environmental conditions. In the GNSS-denied environments such as indoors, subterranean tunnels, and underground mines, their advantages are difficult to exploit

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due to signal blockage [5]. To address this issue, researchers have developed alternative technologies and methods such as those based on inertial measurement unit (IMU) [6], camera [7], [8], light detection and range (LiDAR) [9], Bluetooth [10], ultra-wideband (UWB) [11], and other sensors. Among these various navigation solutions, the inertial navigation system (INS) is commonly used to provide high-frequency estimates of velocity, position, and attitude. However, the accuracy of INS navigation can deteriorate significantly over time due to the unbounded error growth. Visual [12] and LiDAR [13] simultaneous localization and mapping (SLAM) have made significant progress in recent decades. Although a camera is cost-effective, it remains susceptible to variations in brightness and non-textured environments. In contrast, LiDAR can provide range measurements and is invulnerable to illumination variations, which is promising for practical application. Additionally, the cost of LiDAR hardware has been decreasing with technological advancements, making it more accessible to civil and commercial fields [14].

LiDAR has high measurement accuracy and can be used for real-time state estimation. The point cloud coordinates are acquired by propagating laser beams at the speed of light. As the beams encounter materials with diverse compositions and properties, variations in the uncertainty of point cloud measurements occur, resulting in changes in the weighting of their combination with other sensors. Therefore, quantifying the confidence level of laser measurements is crucial for effective multisensor fusion. In SUMA [15], the surfel from point cloud was modeled as a (elliptical) circular plane, comprising radius, position, and normal. Radius was dynamically updated during the map update process to eliminate unstable surfels, and its magnitude was proportional to the measurement accuracy. In [16], the confidence level of the current match was updated using a probabilistic update approach. It proposed a probabilistic surfel map fusion method to represent dense maps, and employed two different map fusion techniques to integrate point clouds from multiple perspectives. M-LOAM [17] incorporated map uncertainty awareness during multi-LiDAR mapping, which ensured map consistency and robustness. It accounted for the propagation of measurement noise and uncertainty caused by extrinsic disturbances within the system. In [18], uncertainties in pose estimation were also introduced into state estimator. And the solving process was similar to normal distribution transform (NDT) [19], where the map was partitioned into independent adaptive probabilistic voxels. The aforementioned methods were primarily designed for pure LiDAR odometry. IMU can be effectively integrated with LiDAR to achieve point cloud motion compensation [20] and enable high-frequency state prediction in LINS [21], LIO-SAM [22], and FAST-LIO2 [23], which further improves the accuracy and reliability performance. However, the point cloud observation noises in these LiDAR-inertial odometry (LIO) schemes were treated as the constant value, neglecting real-world conditions and consequently introducing unavoidable errors. MA-LIO [24] was a multiple LIO system. The continuous-time IMU model was employed to estimate the point-wise uncertainty of the point cloud, considering both temporal and range uncertainties. However, the primary focus of this study was to represent uncertainties in homogeneous sensors. In multimodal fusion systems, the adoption of more accurate uncertainty models is crucial for obtaining the observation weights and state covariance.

At the same time, LIO represents a form of local positioning method characterized by the accumulation of errors over time. Error correction can be achieved through the utilization of loop closure and control point constraints [25]. However, this approach is not applicable in non-repetitive visit areas. In indoor localization, utilizing deployed infrastructure can establish global observation constraints. For instance, WiFi [26], pseudolites [27], and UWB [28], [29], as well as other wireless sensor networks, can serve this purpose. The WiFi module is simple to use and exhibits high availability, but it is susceptible to significant signal strength variation due to multipath propagation, resulting in obvious positioning errors. Pseudolites offer high accuracy in indoor environments, but they come with higher deployment costs and present challenges for practical applications. UWB technology has been increasingly applied in positioning within GNSS-denied environments due to its characteristics of high transmission frequency, low power consumption, and low installation cost. The UWB cooperative system consists of anchor and tag nodes, with tag nodes installed in pedestrian or vehicle equipment receiving ranging information from anchor nodes to achieve localization. To improve the accuracy and reliability of UWB localization, researchers have explored combining UWB observations with IMU measurements. Li et al. [30] adopted an extended Kalman filter to achieve a high output frequency of up to 80 Hz and low latency. Qin et al. [31] employed the unscented Kalman filter fusion framework to enhance precision by mitigating non-line-of-sight (NLOS) occlusion errors for pedestrian localization. Hu et al. [32] introduced an adaptive fuzzy factor graph in NLOS environment to deal with various signal information, demonstrating the effectiveness of the optimization method. Additionally, reliability can be further strengthened through the use of other constraints, such as environmental structure [28] and round-trip-time [29]. However, the previous studies did not take into account the possibility of long-term UWB signal loss, and their experimental scenes were often fixed. In more complex scenarios, the performance of UWB localization may become uncertain. Therefore, it is important to address these issues and develop approaches that can handle long-term signal loss and adapt to changing environments.

Building upon above studies, LiDAR/IMU/UWB fusion state estimation exhibits significant advantages. In an indoor multisource fusion system, UWB plays a role similar to GNSS by providing absolute constraints [11]. However, the application environment of UWB determines that its signals are more susceptible to interference compared to GNSS, and its ranging accuracy is significantly lower than that of GNSS carrier-based methods. Marković et al. [33] employed a loosely coupled extended Kalman filter to fuse the localization outcomes of LiDAR SLAM, visual SLAM, and UWB

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odometry. In contrast, our approach utilizes a tightly coupled strategy, leveraging raw measurements to achieve more precise and robust pose estimation. And Nguyen et al. [34] proposed a LiDAR/UWB/IMU factor graph optimization (FGO) fusion method that deployed two LiDARs and four tag nodes. They added UWB ranging observation, IMU preintegration factor, and scan matching factor to the FGO framework using the sliding-window method. However, the system was implemented with three anchor stations. Thereby, the installation of these anchor stations lacked consideration for both geometric configuration and the implications of NLOS errors. Additionally, the optimization-based methods require higher computational time compared to filtering approaches. In contrast, multianchor initialization and uncertainty modeling techniques have been incorporated into our system, which enabling our system to address the impact of NLOS errors and absence of anchor stations in some scenarios. Furthermore, we adopted the iterative error state Kalman filter framework, which reduce the computational cost.

Overall, in terms of continuous state estimation with reliability and accuracy, there are three primary goals that should be considered.

1) The system should be robust and capable of operating effectively in special environments, such as repetitive environments that may lack sufficient geometric features for LiDAR SLAM.
2) The global localization method should be resilient and able to cope with unpredictable signal transmission issues, such as signal blockage, NLOS, and multipath.
3) Real-time positioning performance is crucial for autonomous systems, as this ensures security and stability.

Therefore, we present the UWB/LiDAR/IMU integration scheme for unmanned ground vehicles (UGVs) in our work. And the innovations are as follows.

1) To the best of our knowledge, it is the first work that employs an iterative error state Kalman filtering (IESKF) framework to tightly integrate UWB, LiDAR, and INS raw measurements.
2) We formulated uncertainty models for point cloud and UWB ranging measurements. The incorporation of the uncertainty-aware module can provide accurate covariance in observation information, which enables the multimodal state estimation system to achieve reliable estimates.
3) We conducted extensive experiments in diverse scenarios. In outdoor experiment, the translation and rotation error have been evaluated. The ablation experiments validate the performance of this algorithm, especially in the anchor-less environment. The indoor experiments confirm that our system can achieve excellent performance in degraded and NLOS scenarios. Moreover, the system has high real-time performance.

II. System Overview

The pipeline of the proposed UWB/LiDAR/INS tightly coupled system is shown in Fig. 1. UWB inputs are fed into the anchor selection module, which aids in the initialization of position and attitude for INS. In our implementation, we calibrate the position of UWB anchor station in the global coordinate system. And we adopt the INS propagation algorithm for high-frequency pose estimation, which is then applied to the motion compensation and outlier removal. Then, we establish uncertainty models based on the measurement principles of the sensors to obtain accurate covariance estimation, facilitating subsequent fusion and processing. The uncertainty in LiDAR is primarily characterized by depth and bearing information, while in the case of UWB information, it is mainly attributed to ranging errors. Finally, an IESKF framework is applied for pose estimator integrating the UWB, LiDAR, and INS raw measurements. And the local map is dynamically updated to ensure consistent localization and mapping.

III. Methodology

A. Notations and State Description

In this section, some parameter notations in our article are listed in Nomenclature section.

Four reference frames are considered in this article, including global frame $\{W\}$, UWB frame $\{U\}$, LiDAR frame $\{L\}$, and inertial frame $\{I\}$. For the sake of convenience, this article refers to $\{I\}$ as the body frame. The parameters to be estimated on manifold $\mathcal{M}$ is as follows:

$$\begin{aligned} &\mathcal{M} \triangleq SO(3) \times \mathbb{R}^{15}; \dim(\mathcal{M}) = 18 \\ &x_{I_k} = \left[R^W_{I_k}, P^W_{I_k}, v^W_{I_k}, b_w, b_a, g^W \right] \in \mathcal{M} \end{aligned} \tag{1}$$

where x_k vector is the IMU state in t_k of orientation $R^W_{I_k}$, position $P^W_{I_k}$, velocity $v^W_{I_k}$, gyroscope bias b_w, and accelerometer bias b_a. And g^W represents the gravity vector. Superscript $\{W\}$ indicates that the current state is expressed in the world frame.

In this article, $R^A_B \in SO(3)$ and $t^A_B \in \mathbb{R}^3$ are the orientation and position part in transform matrix $T^A_B \in SE(3)$, which is denoted as

$$T^A_B = \begin{bmatrix} R^A_B & t^A_B \\ 0^T & 1 \end{bmatrix} \in \mathbb{R}^{4*4}. \tag{2}$$

B. Initialization

The initialization process enables spatial alignment of all sensor coordinates. Moreover, a good initial value facilitates the convergence of state estimation. Current studies primarily focus on initialization using a limited number of anchor stations and in line-of-sight (LOS) scenarios. However, when multiple anchor stations are deployed, NLOS errors will be introduced during the initialization process. As shown in Fig. 2, these errors from yellow line can significantly impact the accuracy of the initialization. In this article, we utilize the traversal-based method for selecting anchor stations. The range measurements are sorted in ascending order, and initial position is computed by sequentially selecting N ranging values. For ith anchor station, the measured distance to the tag node m is denoted as $d_{i,m}$. The true distance (i.e., true state p_m together with anchor point location $P^W_{\text{anchor},\,i} \in \mathbb{R}^2$)

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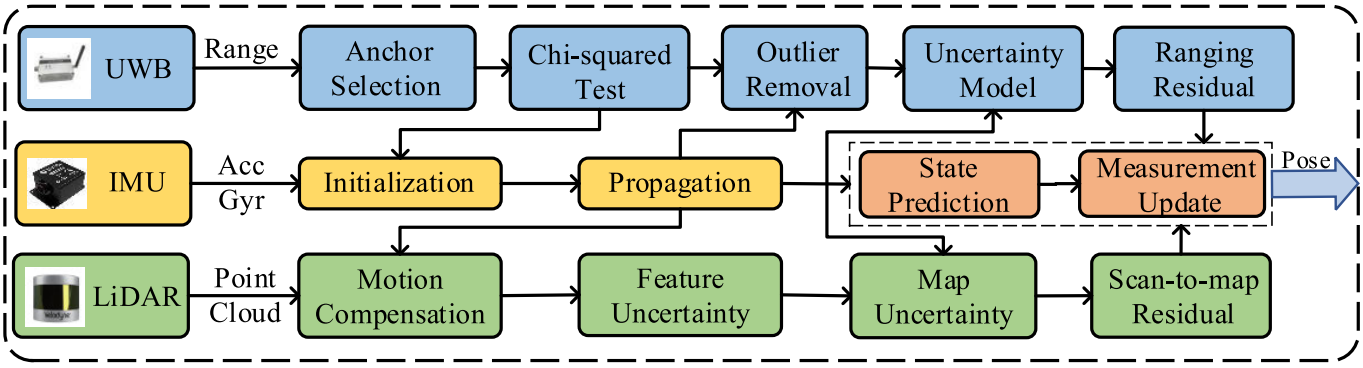


Fig. 1. System overview of the tightly coupled UWB/LiDAR/INS system. We fully utilize the raw measurements of sensors.

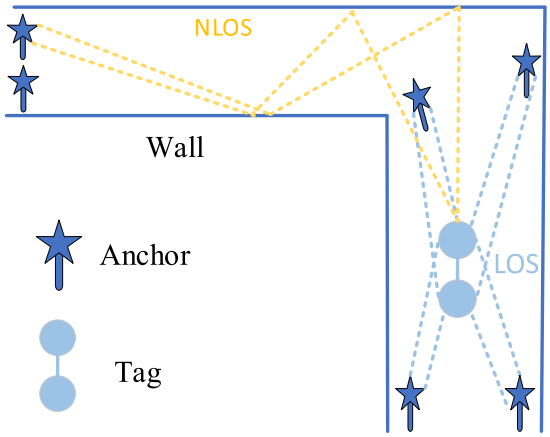


Fig. 2. Distance measurement signals of UWB, where the yellow line represents NLOS measurements, and the blue line represents LOS measurements.

and measured distance lead to zero residual. Now considering the noise, we have

$$\begin{aligned} 0 &= d_{i,m} - \left(\left\| p_m - P^W_{\text{anchor},\,i} \right\| \right) \\ &\approx d_{i,m} - \left(\left\| \bar{p}_m + \Delta p_m - P^W_{\text{anchor},\,i} \right\| \right) + n_{i,m} \end{aligned} \tag{3}$$

where $\bar{p}_m$ and Δp_m are the current updated and error state. Therefore, the least squares problem about Δp_m is

$$\min f\left(\Delta p_m\right) \quad \text{s.t.} \Delta p_m \in \mathbb{R}^2 \tag{4}$$

$$f\left(\Delta p_m\right) = \sum_{i=1}^{N} \left(\left\| d_{i,m} - \left(\bar{p}_m + \Delta p_m - P^W_{\text{anchor},\,i} \right) \right\|^2 \right). \tag{5}$$

Gauss–Newton algorithm [35] is used for the solution and the optimal solution is

$$\Delta p_m = \left(A^T A \right)^{-1} A^T L \tag{6}$$

where

$$A = \begin{bmatrix} \frac{-\left(P^W_{\text{anchor},1} - \bar{p}_m\right)}{\left\| P^W_{\text{anchor},1} - \bar{p}_m \right\|} \\ \vdots \\ \frac{-\left(P^W_{\text{anchor},i} - \bar{p}_m\right)}{\left\| P_{\text{anchor},i} - \bar{p}_m \right\|} \end{bmatrix} \in \mathbb{R}^{N \times 2}$$

and

$$L = \begin{bmatrix} -\left(\left\| P^W_{\text{anchor},1} - \bar{p}_m \right\| - d_{1,m} \right) \\ \vdots \\ -\left(\left\| P^W_{\text{anchor},i} - \bar{p}_m \right\| - d_{i,m} \right) \end{bmatrix} \in \mathbb{R}^{N \times 1}.$$

Typically, the value of N is set to 4 in our implementation. And the updated state is

$$\bar{p}_m = \bar{p}_m + \Delta p_m \tag{7}$$

when $\|\Delta p_m\| \leq \varepsilon$, the iteration process has converged. In the meantime, $\bar{p}_m$ needs to satisfy the chi-square test with the following equation:

$$\chi^2 = \frac{\left\| d_{i,m} - \left(\bar{p}_m - P^W_{\text{anchor},\,i} \right) \right\|^2}{\sigma^2} \tag{8}$$

where σ represents the standard deviation of ranging measurements. When the range measurements from all four anchor stations are in LOS (blue line), the variable χ^2 follows a Gaussian distribution with a mean of zero. By considering this, we can select measurements with good ranging quality.

Ranging observations are used to initialize the position in the horizontal direction. However, calculating vertical elevation will introduce singular values due to the anchor stations being at the nearly equal height in practice. Therefore, we employ the inverse distance weighting (IDW) [36] to initialize the elevation. This method utilizes the elevation values of the N selected anchor stations and performs spatial interpolation to estimate the height of the tag node. The inverse of the distance to the tag node (i.e., $1/d_{im}$) is used as the weight for corresponding interpolation.

Moreover, zero-velocity detection is applied to obtain the rough roll and pitch angles, via keeping device platform static for several seconds (set to 10 s). Additionally, two UWBs are utilized to calculate the initial position in the global coordinate system, which helps to determine the global yaw angle, similar to GNSS dual-antenna. Through the above steps, the initial alignment of the inertial navigation is completed.

C. IMU Model

The primary function of IMU is to update the system state through IMU propagation. It provides high precision in

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short-term estimation and enables the removal of outliers in observations. The measurement models are as follows [23]:

$$\begin{aligned}\dot{R}_{I_k}^W &= R_{I_k}^W \left[\tilde{w}^I - b_w - n_w\right]_\times \\ \dot{P}_{I_k}^W &= v_{I_k}^W \\ \dot{v}_{I_k}^W &= g^W + R_{I_k}^W \left(\tilde{a}^I - b_a - n_a\right) \\ \dot{b}_w &= n_{bw} \\ \dot{b}_a &= n_{ba} \\ \dot{g}^W &= 0\end{aligned} \tag{9}$$

where $[\cdot]_\times$ represents vectors map to a skew symmetric matrix. $\tilde{w}^I$ and $\tilde{a}^I$ denote the gyroscope and accelerometer outputs, respectively, which include the measurements white noise n_w and n_a. Simultaneously, the biases of raw IMU measurements b_w and b_a are modeled as random walk process, which are driven by Gaussian noise n_{bw} and n_{ba}.

Considering the noise w_{I_k}, IMU input u_{I_k}, and kinematic function f, the discrete model between kth and $(k+1)$th IMU measurement can be derived from continuous time model as below

$$\hat{x}_{I_{k+1}} = \hat{x}_{I_k} \boxplus \left(\Delta t f\left(\hat{x}_{I_k}, \mathrm{u}_{I_k}, \mathrm{w}_{I_k}\right)\right) \tag{10}$$

where

$$\begin{aligned}&\mathrm{f}\left(\hat{x}_{I_k}, \mathrm{u}_{I_k}, \mathrm{w}_{I_k}\right) \\ &= \begin{cases} \tilde{w}^I - b_w - n_w \\ v_{I_k}^W + \dfrac{1}{2}\Delta t\left(g^W + R_{I_k}^W\left(\tilde{a}^I - b_a - n_a\right)\right) \\ g^W + R_{I_k}^W\left(\tilde{a}^I - b_a - n_a\right) \\ n_{bw} \\ n_{ba} \\ 0 \end{cases}\end{aligned} \tag{11}$$

and Δt is the interval time for IMU sampling.

D. LiDAR Uncertainty Model

The laser scanner measures depth as well as horizontal and vertical scanning angles. Commonly, we can obtain the 3-D coordinates of each point in the laser scanner coordinate system from the point cloud. It is achieved by utilizing the polar coordinate representation and subsequently converting the spherical coordinates to X–Y–Z coordinates as follows:

$$p_i^L = \lambda_{pi}^L \phi_{pi}^L. \tag{12}$$

And the measurements of depth and bearing are subject to noise due to the computation of laser time-of-flight and the scanning motor encoders. Inspired by Yuan et al. [37], we modeled the uncertainty of points based on depth $\tilde{\lambda}_{pi}^L \in \mathbb{R}^1$ and bearing $\tilde{\phi}_{pi}^L \in \mathbb{S}^2$ measurement. We have

$$\begin{cases} \lambda_{pi}^L = \tilde{\lambda}_{pi}^L + n_{\lambda_{pi}^L} \\ \phi_{pi}^L = \tilde{\phi}_{pi}^L \oplus n_{\phi_{pi}^L} \end{cases} \tag{13}$$

where $n_{\lambda_{pi}^L} \in \mathbb{R}^1$ represents the error in 1-D depth measurement. $\oplus$ denotes that rotating a small perturbation vector $n_{\phi_{pi}^L} \in \mathbb{R}^1$ in the tangent plane with $\tilde{\phi}_{pi}^L$. According to (12) and (13), the point can be modeled as

$$p_i^L = \tilde{\lambda}_{pi}^L \tilde{\phi}_{pi}^L + J_\lambda n_{\lambda_{pi}^L} + J_\phi n_{\phi i}^L \tag{14}$$

where $(J_\lambda n_{\lambda_{pi}^L} + J_\phi n_{\phi_{pi}^L}) \in \mathcal{N}(0, \Sigma_{p_i^L})$ represents the measurement noise. The detailed calculation process is presented in the Appendix. Therefore, the covariance matrix can be derived as

$$\Sigma_{p_i^L} = J_\lambda n_{\lambda_{pi}^L} (J_\lambda)^T + J_\phi n_{\phi_{pi}^L} \left(J_\phi\right)^T. \tag{15}$$

E. Ranging Uncertainty Model

UWB provides distance measurements between anchor stations and tags, and the uncertainty is determined by the range measurement error. We focus on the planar constrains from the raw range information. The anchor stations in the frame $\{W\}$ is $P_{\text{anchor},i}^W \in \mathbb{R}^2, i \in \{1, 2, 3 \ldots, N\}$. The measurement model is defined as below

$$d_{i,m_j} = \sqrt{\left\| P_{\text{anchor},i}^W - \hat{P}_{\text{tag},j}^W \right\|^2 + \left(\Delta h_{i,m_j}\right)^2} + n_{i,m_j} \tag{16}$$

$$\hat{P}_{\text{tag},j}^W = \left(\hat{P}^W + \hat{R}^W L_{U,m_j}^I\right) \tag{17}$$

where $\hat{R}^W \in SO(2)$ and $\hat{P}^W \in \mathbb{R}^2$ are from the state of IMU propagation. In our implementation, $m_j, j \in \{1, 2\}$ represents the tag node. $L_{U,j}^I \in \mathbb{R}^2$ denotes the planar lever arm from the corresponding fixed tag to the IMU. Hence, $\hat{P}_{\text{tag},j}^W$ is the prediction UWB tag position from IMU propagation. $\Delta h_{i,m_j}$ is the height difference between ith anchor and jth tag. And n_{i,m_j} represents the measurement noise.

For wireless communication systems, signal transmission can be obstructed by various factors such as vehicles, pedestrians, buildings, and other obstacles. It leads to multipath and NLOS errors, which can significantly degrade the performance of the system. Due to the significant ranging errors introduced by NLOS anchor stations, a residual test is employed to mitigate their impact

$$\left|n_{i,m_j}\right| < \vartheta \tag{18}$$

and ϑ is the threshold for ranging residual. The empirical value is set to 0.25 m. We utilize the property of high short-term accuracy in the IMU propagation results to eliminate the errors in range measurements. When noise is significantly higher, the measured ranging d_{i,m_j} will be discarded.

F. State Estimation

According to the aforementioned sensor model, state estimation can be achieved by incorporating IMU measurement, point cloud observations, and UWB ranging measurements. Therefore, IESKF consists of the following steps.

1) State Prediction: In the IMU prediction process, setting the noise w_{I_k} to zero, the propagation of state $\hat{x}_{I_k}$ is defined as

$$\hat{x}_{I_{k+1}} = \hat{x}_{I_k} \boxplus \left(\Delta t \mathrm{f}\left(\hat{x}_{I_k}, \mathrm{u}_{I_k}, 0\right)\right). \tag{19}$$

The prediction of error state would be

$$\delta_{\hat{x}_{I_{k+1}}} = x_{I_{k+1}} \boxminus \hat{x}_{I_{k+1}} = F_{\delta\hat{x}_k} \delta\hat{x}_{I_k} + F_{\mathrm{w}_k} \mathrm{w}_{I_k} \tag{20}$$

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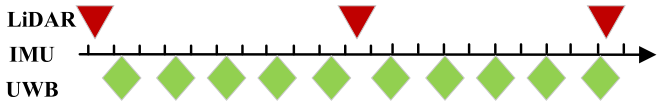


Fig. 3. Frequencies of LiDAR, IMU, and UWB are 10, 100, and 50 Hz, respectively.

which is presented in the Appendix. $F_{\delta\hat{x}_k}$ and F_{w_k} are the partial differentiation with respect to error state $\delta_{\hat{x}_{Ik+1}}$ under the condition when $\delta\hat{x}_{I_k} = 0$ and $\mathrm{w}_{I_k} = 0$ [23]. Therefore, the noise covariance matrix is

$$\hat{\Sigma}_{k+1} = F_{\delta\hat{x}_k}\hat{\Sigma}_k F_{\delta\hat{x}_k}^T + F_{\mathrm{w}_k} Q F_{\mathrm{w}_k}^T \tag{21}$$

where Q represents the covariance of noise w_{I_k}.

As shown in Fig. 3, the IMU propagation process occurs after the LiDAR/UWB measurement update. IMU prediction provides high-frequency pose information for point cloud motion compensation and ranging outlier removal. Each IMU measurement is updated using the closest timestamped observation, and the state is interpolated to achieve updates.

2) LiDAR Observation: IMU prediction is used for propagating the uncertainty in the point cloud. And the points in the $\{L\}$ frame are transformed to $\{W\}$ frame. $T_L^I = (R_L^I, L_L^I)$ is the extrinsic matrix. We have

$$p_{L_i}^W = \hat{R}_{I_k}^W \left(R_L^I p_i^L + L_L^I\right) + \hat{P}_{I_k}^W. \tag{22}$$

$\hat{T}_{I_k}^W = (\hat{R}_{I_k}^W, \hat{P}_{I_k}^W)$ is the IMU prediction state. This process also introduces the estimation pose uncertainty $(\Sigma_{\hat{R}_{I_k}^W}, \Sigma_{\hat{P}_{I_k}^W})$, combining (14) and (22), the point covariance in $\{W\}$ frame can be formulated

$$\Sigma_{p_i^W} = \hat{R}_{I_k}^W \Sigma_{p_i^L} \left(\hat{R}_{I_k}^W\right)^T + \Sigma_{\hat{P}_{I_k}^W} + J_R^W \Sigma_{\hat{R}_I^W} \left(J_R^W\right)^T \tag{23}$$

where $J_R^W = \hat{R}_I^W [R_L^I p_i^L + L_L^I]_\times$. As for the plane feature, it consists of points. Hence, the mean point and covariance matrix are as follows:

$$\bar{p} = \frac{1}{y}\sum_{i=1}^{y} p_{L_i}^W;\ A = \frac{1}{y}\sum_{i=1}^{y}\left(p_{L_i}^W - \bar{p}\right)\left(p_{L_i}^W - \bar{p}\right)^T. \tag{24}$$

The eigenvector corresponding to the smallest eigenvalue of the plane covariance matrix represents the normal vector μ of the plane. Thus, the plane feature can be represented by the normal vector μ and the mean point $q_i^W = \bar{p}$. The uncertainty of each point jointly contributes to the uncertainty of the plane. The plane covariance matrix is

$$\Sigma_{\mathrm{plane}} = \sum_{i=1}^{y} \frac{\partial f}{\partial p_{L_i}^W} \Sigma_{p_i^W} \left(\frac{\partial f}{\partial p_{L_i}^W}\right)^T. \tag{25}$$

$(\partial f / \partial p_{L_i}^W)$ is the derivative of normal vector μ and the mean point q_i^W with respect to $p_{L_i}^W$. The detail computation can be referred to [18] and [38].

Once a scan of LiDAR data is received, it is registered with the local map to perform data association. LiDAR point measurement model in the scan-to-map module is as follows:

$$r_L\left(x_k, p_i^L\right) = \mu^T\left(p_{L_i}^W - q_i^W\right). \tag{26}$$

where $p_{L_i}^W$ represents the current LiDAR point in the $\{W\}$ frame. r_L is the point-to-plane distance residual, which should converge to zero when there is no noise. Therefore, considering the point noise, the observed equation can be modeled as

$$\begin{aligned} 0 = r_L\left(x_k, p_i^L\right) \approx r_L\left(\hat{x}_k, p_i^L\right) + H_{p_i}^L \delta_{\hat{x}_{I_k}} \\ + J_\mu n_\mu + J_{q_i^W} n_{q_i^W} + J_{p^L} n_{p^L}. \end{aligned} \tag{27}$$

And the derivative with respect to normal vector and mean point can be derived from (22) and (26)

$$J_u = \hat{T}_{I_k}^W T_L^I p_i^L - q_i^W \tag{28}$$

$$J_{q_i^W} = -\mu^T \tag{29}$$

$$J_{p^L} = \mu^T \hat{R}_{I_k}^W R_L^I. \tag{30}$$

Assuming that $\alpha = J_\mu n_\mu + J_{q_i^W} n_{q_i^W} + J_{p^L} n_{p^L}$ is the uncertainties of all errors, the Jacobian of error is defined as

$$J_\alpha = \begin{bmatrix} J_\mu & J_{q_i^W} & J_{p^L} \end{bmatrix}. \tag{31}$$

And we have the Jacobian matrix $H_{p_i}^L$

$$H_{p_i}^L = \mu^T \begin{bmatrix} -\hat{R}_{I_k}^W \left[R_L^I p_i^L + L_L^I\right]_\times & I_{3\times 3} & 0_{3\times 12} \end{bmatrix}. \tag{32}$$

Combining (15) and (25), the covariance matrix of LiDAR observations is

$$\Sigma_\alpha = J_\alpha \begin{bmatrix} \Sigma_{\mathrm{plane}} & 0 \\ 0 & \Sigma_{p_i^L} \end{bmatrix} (J_\alpha)^T. \tag{33}$$

3) Ranging Observation: According to (16) and (17), we formulate the problem about ranging measurements. Let $\tilde{d}_{i,m_j}$ be the 1-D ranging value with an error term n_u

$$\tilde{d}_{i,m_j} = d_{i,m_j} + n_u. \tag{34}$$

Therefore, the kth UWB measurement between ith anchor and jth tag is used for updating the current state $\hat{x}_{I_k}$ and the residual is

$$\begin{aligned} 0 = r_u\left(x_k, d_{i,m_j}\right) \approx r_u\left(\hat{x}_k, d_{i,m_j}\right) + H_{i,j}^u \delta_{\hat{x}_{I_k}} + \beta \\ \beta \in \mathbb{N}\left(0.\Sigma_\beta\right). \end{aligned} \tag{35}$$

Σ_β denotes the covariance about the measurement noise. And the Jacobian matrix $H_{i,j}^u$ is

$$H_{i,j}^u = \begin{bmatrix} 0_{3\times 1} & -\frac{a}{d_{i,m_j} - \sqrt{\|a\|^2 + \left(\Delta h_{i,m_j}\right)^2}} & 0_{13\times 1} \end{bmatrix}. \tag{36}$$

In our deduction, $a = (\hat{P}^W + \hat{R}^W L_{U,m_j}^I) \in \mathbb{R}^2$ is the same with $\hat{P}_{\mathrm{tag},j}^W$ in (17).

4) Update of Filtering: Combining the prior and observation residuals from the previous steps, we have the maximum a posteriori (MAP) problem

$$\begin{aligned} \min\Bigg(\left\| x_{I_k} \boxminus \hat{x}_{I_k} \right\|_{\hat{\Sigma}_k}^2 + \sum_{i=1}^{y} \left\| r_L\left(x_k, p_i^L\right) \right\|_{\Sigma_\alpha}^2 \\ + \sum_{i=1}^{N}\sum_{j=1}^{M} \left\| r_u\left(x_k, d_{i,j}\right) \right\|_{\Sigma_\beta}^2 \Bigg). \end{aligned} \tag{37}$$

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The IESKF is applied to solve this problem. Hence, observation vector is

$$Z_k = \Big[r_L\left(x_k, p_i^L\right) \ldots, r_L\left(x_k, p_y^L\right), r_u\left(x_k, d_{1,1}\right), \ldots r_u\left(x_k, d_{N,1}\right), \ldots r_u\left(x_k, d_{1,M}\right), \ldots, r_u\left(x_k, d_{N,M}\right) \Big]^T \tag{38}$$

and we have measurement matrix H and noise matrix R is as follows:

$$H = \left[H_{p_1}^L, \ldots H_{p_y}^L, H_{1,1}^u, \ldots H_{N,1}^u, \ldots H_{1,M}^u, \ldots H_{N,M}^u \right]^T \tag{39}$$

$$R = \text{diag}\left(\Sigma_\alpha, \ldots, \Sigma_{\alpha_y}, \Sigma_{\beta_{1,1}}, \ldots \Sigma_{\beta_{N,1}}, \ldots \Sigma_{\beta_{1,M}}, \ldots \Sigma_{\beta_{N,M}}\right). \tag{40}$$

In the $(\kappa+1)$th iteration, state is updated by

$$\hat{x}_{I_k}^{\kappa+1} = \hat{x}_{I_k}^{\kappa} \boxplus \left(-K Z_k - (I - KH)\left(J^\kappa\right)^{-1} \left(x_{I_k}^\kappa \boxminus \hat{x}_{I_k} \right) \right) \tag{41}$$

where J^κ is the Jacobian matrix of $x_{I_{k+1}} \boxminus \hat{x}_{I_{k+1}}$ with respect to $x_{I_k}^\kappa$. The Kalman gain K is

$$K = \left(H^T R^{-1} H + P^{-1} \right)^{-1} H^T R^{-1} \tag{42}$$

and $P = J^\kappa \hat{\Sigma}_k \left(J^\kappa\right)^T$. After convergence of the iterations (i.e., $\hat{x}_{I_k}^{\kappa+1} - \hat{x}_{I_k}^{\kappa} < \xi$), we have the optimal state estimate $\bar{x}_{I_k} = \hat{x}^{\kappa+1}$ and $\bar{\Sigma}_{I_k} = (I - KH)P$. Afterward, we utilize the obtained results to perform the updating of local map and initiate the next cycle of propagation.

IV. Experimental Evaluation

A. Device

The device packed in the UGV is depicted in Fig. 4(b), which includes two Nooploop LinkTrack P-B UWBs, an ADIS16497 IMU, a Velodyne VLP-16 LiDAR, and the GNSS dual-antenna. RTK/INS-integrated navigation was the benchmark for evaluation in outdoor environment, and RTK corrections were from China Mobile base station. The RTK/INS-integrated navigation data was post-processed using NovAtel Inertial Explorer 8.9 commercial software. We chose to conduct precision validation experiments in an outdoor environment in Sections IV-B and IV-C due to the availability of high-precision dynamic benchmarks. The spatial position among all sensors has been measured accurately. Meanwhile, the industrial camera was only used to record data, which was not involved in our algorithm. UWB anchor stations were calibrated according to CHCNAV GNSS geodetic receiver (localization accuracy 2.5 mm + 0.5 ppm) in Fig. 4(a). Tag node has the cap antenna due to the larger signal reception area as shown in Fig. 4(c).

B. Local Performance

In this section, the relative trajectory error is used for evaluating the performance of local positioning (LIO system). In order to evaluate the continuous accuracy of our system, we conducted experiments in an environment with considerable satellite visibility and employed RTK/INS navigation, which provided centimeter-level precision. Three typical scenarios were selected in campus, which are listed in Table I.

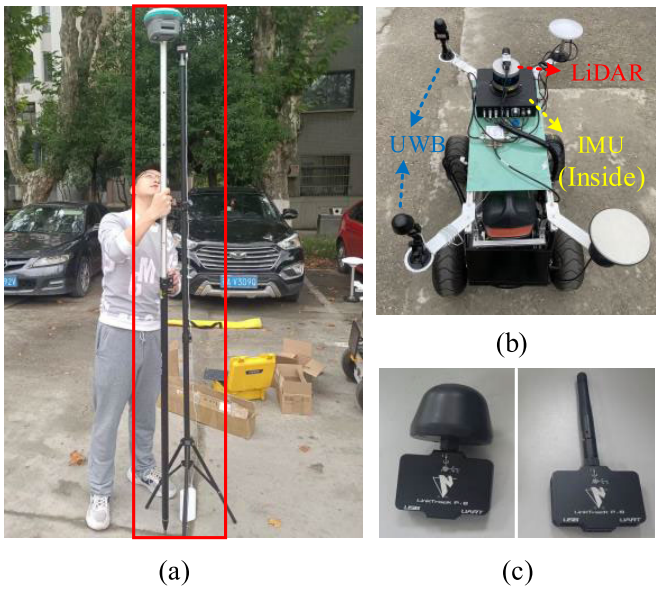


Fig. 4. Experimental equipments. (a) Calibrating the anchor station via CHCNAV GNSS geodetic receiver. (b) UGV platform with sensors. (c) Left side represents the tag node, while the right side is the anchor node.

TABLE I
Details of Three Scenarios

Dataset	Length(m)	Characteristics
dataset1	274.63	Sharp turns
dataset2	249.10	Open area, dynamic objects
dataset3	257.83	Straight line, flat ground

Relative translation error (RTE) and relative rotation error (RRE) per 100 m are chosen to evaluate the local state estimation performance. And root-mean-square error (RMSE) is used to evaluate the accuracy. We choose open-source state-of-the-art LiDAR SLAM approaches such as LIO-SAM [22],[1] a tightly coupled, smoothing optimization system, LiLi-OM [39],[2] a hierarchical sliding window, direct LiDAR-inertial method, and FAST-LIO2 [23],[3] a direct and fast odometry. All parameters are the recommended value in their own algorithms for outdoor environment. And our method utilizes the same parameters all the time without loop closure module.

We carry out the experiments by preventing the UWB information from the Kalman filtering. Positioning results are shown in Table II.

As can be seen, our LiDAR/IMU odometry has the best performance compared with other LiDAR-inertial SLAM benchmarks. It can be attributed to the uncertainty model in scan-to-map and filtering architecture, which enhances the precision and efficiency. In dataset1, sharp-turn motion severely degrades the feature tracking performance, and it seems that the positioning accuracy of other systems is poor. However,

[1]https://github.com/TixiaoShan/LIO-SAM
[2]https://github.com/KIT-ISAS/lili-om
[3]https://github.com/hku-mars/FAST_LIO

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TABLE II
LOCAL PERFORMANCE OF POSITION AND ATTITUDE

Scheme	RTE (%)			RRE (deg)		
	dataset1	dataset2	dataset3	dataset1	dataset2	dataset3
LIO-SAM	1.10	1.17	2.44	0.73	0.82	0.83
LiLi-OM	4.72	Fail	2.64	2.82	Fail	7.88
FAST-LIO2	3.36	2.07	4.95	2.31	**0.61**	2.00
LiDAR/INS(ours)	**0.82**	**0.99**	**0.65**	**0.71**	0.73	**0.73**

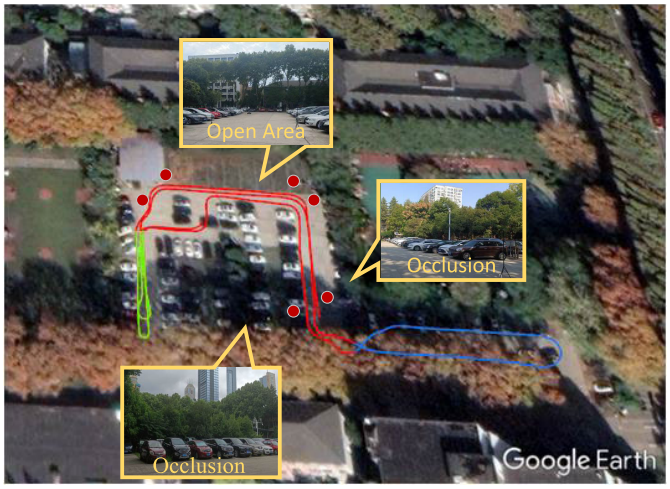


Fig. 5. Trajectory in the experiment. The green region receives signals from two anchor stations, while the blue region does not have any ranging signal. The red points on the map represent the locations of the anchor stations.

our system maintains high precision. We note that LiLi-OM is hard to exert the advantages in dataset2. The reason is due to the INS pre-integration parameters, which need tuned values for different quality IMU. Also, accurate front-end odometry estimation is critical for effective back-end optimization in LiLi-OM, as feature-less open area and dynamic objects can cause feature matching errors in front-end. Simultaneously, this has a minor impact on the uncertainty module, leading to slightly higher rotational errors in our system compared to FAST-LIO2. In dataset3, the scene is simpler than the other two, and our system continues to maintain better performance.

C. Global Performance and Ablation Study

The integration of UWB can minimize drift in local positioning methods and enable global position and orientation estimation. Accordingly, we perform an analysis of translation errors in the east (E), north (N), and up (U) directions, as well as attitude errors in roll, pitch, and yaw. The trajectory covered a length of 1212.63 m and lasted for a duration of 725 s. It encompassed areas with both good signal reception from the anchor stations and obstructed regions as shown in Fig. 5. Since [33] and [34] are not open-sourced, we conducted ablation experiments to validate the system's performance. There are five methods in the comparative analysis.

1) *UWB:* UWB positioning method based on range measurements using extended Kalman filtering (EKF).
2) *UWB/INS [30][4]:* UWB and INS fusion approach.

[4]https://github.com/lijx10/uwb-localization

TABLE III
GLOBAL PERFORMANCE OF POSITION AND ATTITUDE

Scheme	Position (m)			Attitude (deg)		
	E	N	U	Roll	Pitch	Yaw
UWB	0.35	0.37	\	\	\	\
UWB/INS	0.19	0.22	\	0.77	0.75	2.30
LiDAR/INS	0.15	0.07	0.09	1.65	2.09	**1.43**
ULI-LC	0.23	0.31	**0.04**	1.96	4.16	2.99
ULI-TC	**0.07**	**0.06**	0.06	**0.61**	**0.46**	1.45

3) *LiDAR/INS:* The proposed LiDAR-inertial state estimator. Range information from UWB is solely utilized for initialization and is not integrated into the system thereafter.
4) *ULI-LC:* The loosely coupled UWB/LiDAR/INS approach, which is modified from LIO-SAM [22]. We replace the GNSS global localization factor with UWB odometry factor within the FGO framework in LIO-SAM.
5) *ULI-TC:* The proposed tightly coupled UWB/LiDAR/inertial fusion scheme.

Since all the anchor stations have the same height, the accuracy of up direction estimation in 1) UWB and 2) UWB/INS is extremely poor. We employ the method described in Section III-B to estimate the height. However, the up error will quickly diverge in anchor-less environment. And pure UWB odometry cannot calculate attitude. Therefore, we have not evaluated these items in Table III.

From the table, it can be observed that our tightly coupled system ULI-TC has a significant advantage over the factor graph system ULI-LC, especially in terms of attitude estimation. Indeed, the improved performance of the tightly coupled method is attributed to its ability to fully leverage the capabilities of individual sensors. Additionally, it is worth noting that the iterative Kalman filtering and optimization-based approaches are theoretically equivalent [40]. However, in the ULI-LC method, the poor quality of UWB ranging signals leads to the incorporation of UWB positioning results with significant NLOS errors in the blue region, resulting in substantial errors as depicted in Fig. 6. And the uncertainty of UWB positioning in the loosely coupled system is challenging to assess. When the fusion weight about UWB odometry is too small, it will lead to positioning jumps and significantly affect the estimation of the yaw angle. Furthermore, the ULI-LC system encounters challenges in correcting the roll and pitch errors introduced by the low-beam LiDAR, leading to significant inaccuracies in the estimation of these attitude parameters. This conclusion can be deduced from Fig. 7.

Ablation experiments are also analyzed. In the UWB positioning, when there are significant errors in ranging signals, the positioning is interrupted, resulting in a straight error line in the blue region in Fig. 6. ULI-TC odometry is the promotion based on UWB/INS and LiDAR/INS integration. The UWB/INS system exhibits divergence in anchor-less areas, while the roll and pitch remain acceptable. Conversely, the LiDAR/INS system demonstrates smaller position errors compared to UWB/INS and the best yaw estimation in all

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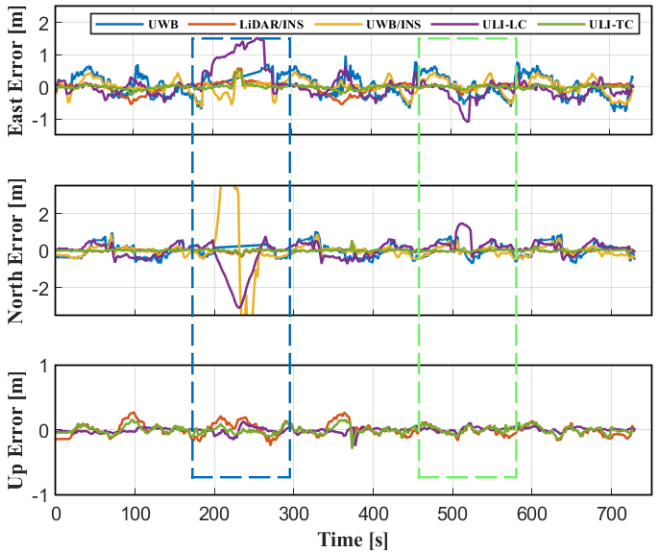


Fig. 6. East, north, and up error in the experiment. The blue and green boxes represent the error values for scenarios with no anchor station and two anchor stations, respectively.

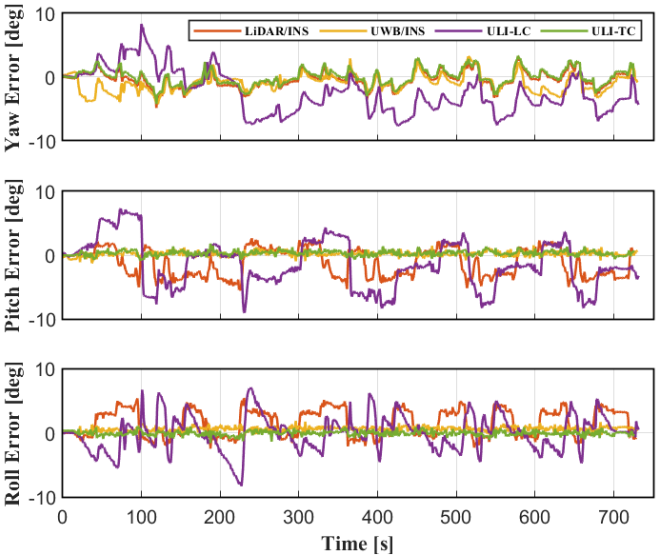


Fig. 7. Attitude difference in yaw, pitch, and roll angles. ULI-LC performs poorly in terms of attitude estimation, and LiDAR/INS has the inaccurate roll and pitch estimation.

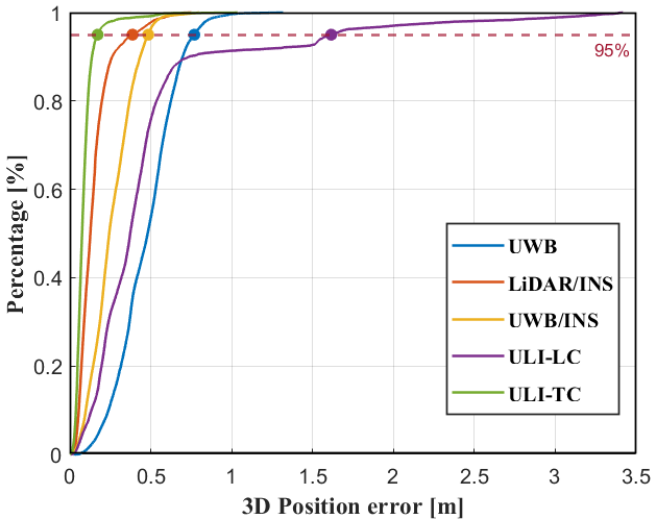


Fig. 8. CDFs of 3-D position errors of UWB, LiDAR/INS, UWB/INS, ULI-LC, and ULI-TC.

TABLE IV
95% CONFIDENCE INTERVAL

	UWB	LiDAR/INS	UWB/INS	ULI-LC	ULI-TC
95% error [m]	0.77	0.39	0.48	1.62	**0.17**



(a) (b)

Fig. 9. (a) Electronic total station. (b) Calibrating the anchor station position via known global point outside.

methods. Optimal performance can be achieved by integrating information from all three sensors in the ULI-TC scheme. The positional errors in the east, north, and up directions are limited to 10 cm, ensuring high accuracy. Moreover, the roll and pitch difference are effectively restricted to less than 0.7°, and the yaw difference is controlled below 1.5°.

The 3-D position errors are further analyzed through cumulative distribution functions (cdfs). As shown in Fig. 8, the line corresponding to 0.95 in the figure represents that 95% of the localization errors fall within these values. The results are summarized in Table IV. In ULI-TC method, 95% of the errors are less than 0.17 m, while for pure UWB positioning, LiDAR/INS, UWB/INS, and ULI-LC method, the values are 0.77, 0.39, 0.48, and 1.62 m, respectively. Therefore, we can conclude that our system has a clear advantage in terms of precision and availability.

D. Indoor Environment

There is no GNSS signal in indoor environment. Hence, in order to obtain global coordinate point, we implemented the FOIF RTS102R10+ Electronic Total Station (ranging accuracy 2 mm + 2 ppm, angular accuracy 2 s) to get the global position of anchor stations and evaluation points. As shown in Fig. 9(a), CHCNAV GNSS geodetic receiver is used for the initial global reference in open area of the roof. Then, several indoor points can be calibrated in Fig. 9(b) according to Electronic Total Station ranging and angle surveying. Two indoor experiments were conducted as follows.

1) *Experiment1:* The experiment was conducted in the Fig. 10, which included both the hall and corridor environments. Four anchor stations were employed, and the

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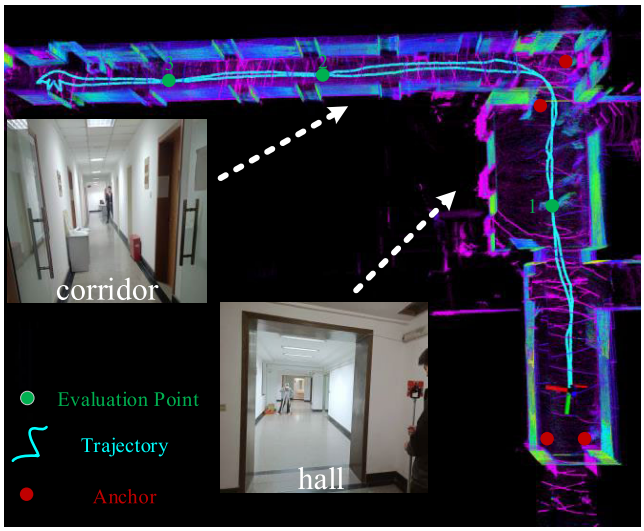


Fig. 10. Mapping result in top view of Experiment1, which includes both hall and corridor scenarios. The red points denote the positions of anchor stations, and the green points represent evaluation locations.

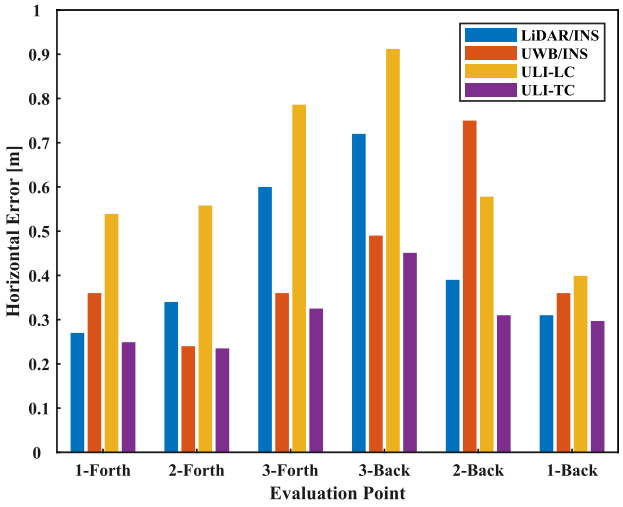


Fig. 11. Detailed horizontal error. Our tightly coupled system has the optimal performance.

experimental trajectory encompassed reciprocal motion along an L-shaped pattern, which covered a distance of approximately 80 m. We also calibrated three points on the ground as the evaluation benchmark. The points in the forward and backward direction have been labeled as (1-forth, 2-forth, 3-forth) and (3-back, 2-back, 1-back) respectively.

2) *Experiment2:* Building on Experiment1, experimental UGV was controlled to depart from and return to the same point and direction. Six anchor stations were used in this implementation. The trajectory formed a loop and covered a total distance of 100 m round trip. The experiment was conducted to validate the system performance in multiscene switching. We evaluated the end-to-end drift using ArUco marker [41].

In Experiment1, Point1 was in the hall, where the anchor stations geometry configuration was better than corridor environment (Point2 and Point3). When UGV was arriving at evaluation point through the use of markers, we remain it stationary for error statistics. Due to the absence of truth value in the height direction, we evaluated the planar 2-D RMSE, which was averaged for 30 s. And the results are shown in Table V.

TABLE V
HORIZONTAL ERROR OF INDOOR EVALUATION POINTS

Point	Horizontal RMSE Error [m]			
	LiDAR/INS	UWB/INS	ULI-LC	ULI-TC
1-forth	0.27	0.36	0.54	**0.25**
2-forth	0.34	0.24	0.56	**0.24**
3-forth	0.60	0.36	0.79	**0.33**
3-back	0.72	0.49	0.91	**0.45**
2-back	0.39	0.75	0.58	**0.31**
1-back	0.31	0.36	0.40	**0.30**

It should be noted that pure UWB odometry introduced in Section IV-C is unable to function in the corridor due to the spatial distribution of the anchor stations. Moreover, LiDAR SLAM methods lead to degeneration in repetitive structural environment, which is difficult for motion estimation convergence in the featureless corridor. The detailed diagram is shown in Fig. 11. In the Point1, the error is obviously smaller than other evaluation points. And ULI-LC is significantly affected by UWB odometry. The large error of LiDAR/INS in the positioning is due to the influence of repetitive features, especially for the Point3. The UWB/INS system performs reasonably well in the area covered by anchor stations, but it introduces significant positioning errors at farther distances from the anchor stations. It is mainly attributed to the decrease in ranging accuracy with increasing distance. Despite these challenges, our scheme is robust enough to handle the scene.

In Experiment2, the mapping results are shown in Fig. 12. The narrow corridor is only 1-m wide, just enough to allow our UGV to pass through. Existing LiDAR SLAM algorithms generally fails in this scenario. LIO-SAM and LiLi-OM both fail in the narrow corridor. The reason is that the front-end feature extraction is out of operation in the degraded environment. It is worth noting that LIO-SAM exhibits jump in the initial moments due to the inability to extract suitable edge features in the wall environment of the hall, which greatly affects the convergence of the back-end state estimation. FAST-LIO2 directly registers current points to the local map, which is more robust in this scene. However, it also proves to drift in the entire trajectory. Meanwhile, the ranging measurements in the narrow corridor and NLOS environment are of poor quality. Therefore, pure UWB-based localization method based on EKF is difficult to achieve success in the environment. However, our system can maintain smooth localization in this challenging scene. The end-to-end drift is 0.03 m in translation and 0.42° in rotation. It is worth noting that the error value in Experiment2 is much smaller than in Experiment1 due to the end-to-end drift, which is typically smaller compared to the process error.

E. Uncertainty-Aware Verification

In this section, we validate the impact of the uncertainty-aware module on the experimental system.

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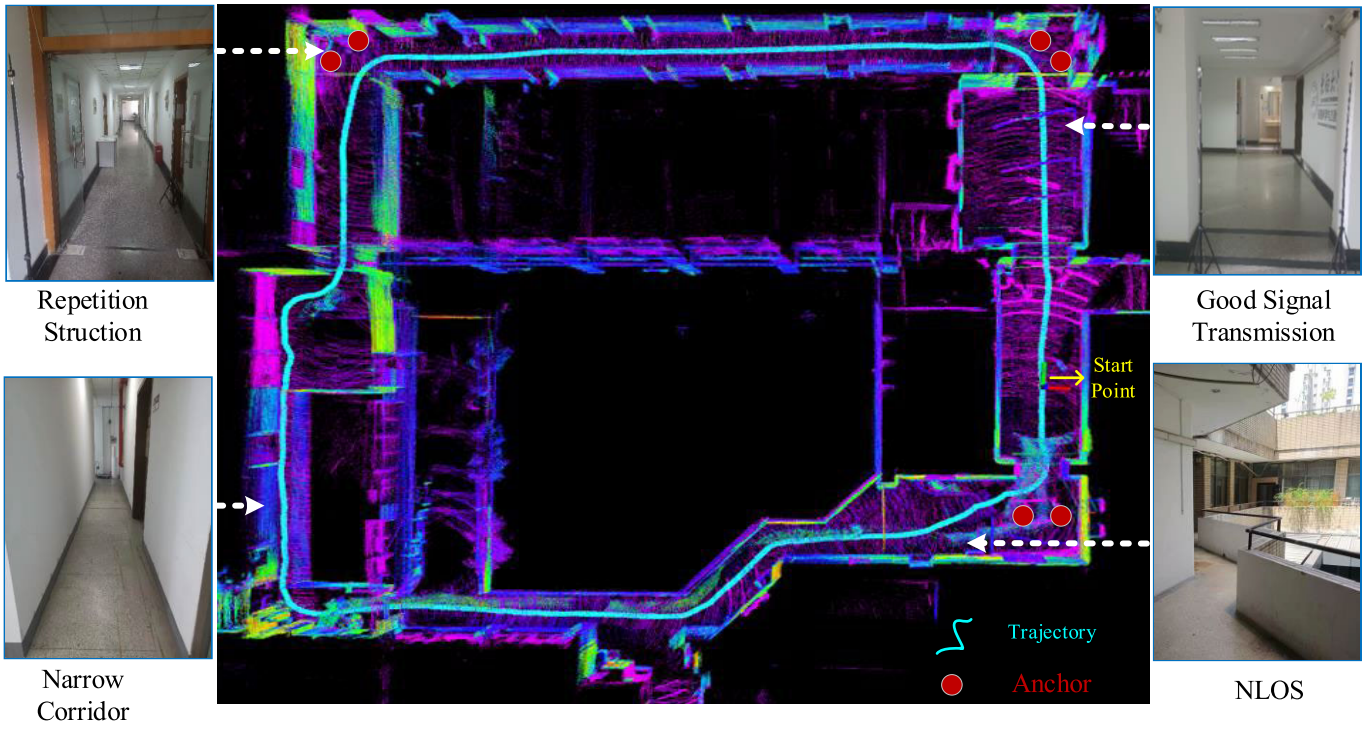


Fig. 12. Mapping result of indoor and degeneration environment in top view of Experiment2. Actual environment is depicted in the photograph. The end-to-end drift is 0.03 m in translation and 0.42° in rotation.

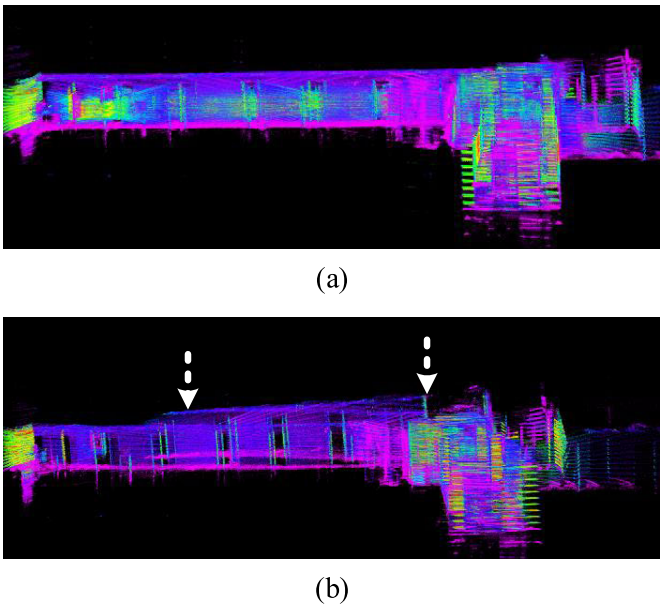


Fig. 13. Side view of the mapping result in Experiment1. (a) With uncertainty-aware module. (b) Ghosting and upward drift.

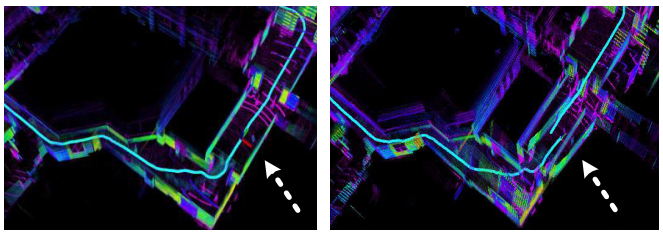


Fig. 14. Magnified view of the endpoint in Experiment2. (a) With uncertainty-aware module. (b) Exhibits obvious drift in the east direction.

We choose indoor environment for analysis, as indoor experiments often include more degraded scenes for LiDAR and NLOS errors for UWB ranging. The accurate estimation of covariance is required in the multisensor fusion system to enhance the reliability of the overall estimation process.

In the comparative experiment, the LiDAR observation convariance matrix (33) is set as a constant value. As shown in Fig. 13, it has ghosting and upward drift between the white arrows without uncertainty-aware module in Experiment1. It is due to the long corridor and insufficient ranging information, which notably affect the accuracy of interframe state estimation. And in Experiment2, the fixed covariance causes imprecise LiDAR observations, while poor geometric configuration of anchor stations in the east direction leads to lower positioning accuracy in that direction. Consequently, this affects the overall accuracy of the tightly coupled filter, which shows obvious drift shown in Fig. 14. Therefore, we can conclude that reliable covariance estimation is the necessary requirement for achieving high-precision localization and mapping results. And our state estimation system incorporates the uncertainty modules can yield better performance in this regard.

F. Computational Complexity

Real-time positioning ability is necessary for unmanned system with autonomous task. In this section, we conduct the computation efficiency evaluation in an Intel i9-11900K

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TABLE VI
AVERAGE TIME OF EACH MODULE

Module	Time Consuming (ms)
Point cloud preprocessing	0.36
UWB Range value preprocessing	0.01
State estimation	3.52
Ins propagation	1.1
Mapping	19.83

TABLE VII
COMPARISON BETWEEN FGO AND IESKF IN TERMS OF TIME CONSUMPTION

	FGO	IESKF
Mean per scan (ms)	62.89	**24.82**

CPU and 32-GB RAM computer. In the outdoor environment, the experiment from Section IV-C is selected for further analysis due to the higher number of point cloud feature points, which requires more computational resources. The average processing time of this outdoor experiment is calculated and presented in Table VI.

Compared with the FGO method, the scan-to-map and mapping module used in ULI-LC can be time-consuming, especially in environments with many features, taking 62.89 ms per scan. However, our scheme can reduce the time consuming to under 25 ms per frame in Table VII due to the IESKF and voxel map storage structure [18], making it suitable for processing LiDAR data at 10 Hz. The preprocessing of UWB involves removing some abnormal ranging values, which only consumes a small amount of time (0.01 ms). It should be noted that INS propagation time also includes the de-skew procedure, which is beneficial to point cloud motion compensation. Furthermore, we have also implemented a multithreaded approach, making our system computationally efficient. In conclusion, the filtering system is more suitable for real-time UWB/LiDAR/INS state estimation and mapping.

V. CONCLUSION AND PROSPECT

In this article, we present an uncertainty-aware-integrated UWB/LiDAR/IMU state estimation framework, including initialization, IMU propagation, establishment of uncertainty model and state estimation with multimodal observations. In outdoor accuracy evaluation, the ULI-TC integration demonstrates superior robustness and precision compared to subsystems and other state-of-the-art algorithms. Moreover, in indoor environments, the tightly coupled system can effectively handle the degradation of feature point clouds and mitigate the impact of NLOS errors in ranging measurements. The incorporation of an uncertainty-aware module can assist the system in accurately constructing current covariance estimates, which enhance the capability of state estimation in degraded and complex environments. We also demonstrate the superior real-time performance of the IESKF in the system. Our work provides a feasible positioning solution for urban underground roadway, shopping mall, or other GNSS-denied environments with localization requirements.

Although the results are encouraging, there is still improvement in the future. Visual semantic features are favorable for odometry and mapping [42], which are also significant for obstacle avoidance and path planning. Therefore, in the next phase of development, camera information will be incorporated into semantic mapping.

APPENDIX

In (14), the point model is defined as

$$\begin{aligned} p_i^L &= \left(\tilde{\lambda}_{pi}^L + n_{\lambda_{pi}^L}\right)\left(\tilde{\phi}_{pi}^L \oplus n_{\phi_{pi}^L}\right) \\ &= \tilde{\lambda}_{pi}^L \tilde{\phi}_{pi}^L + \tilde{\phi}_{pi}^L n_{\lambda_{pi}^L} - \tilde{\lambda}_{pi}^L \left[\tilde{\phi}_{pi}^L\right]_\times N\left(\tilde{\phi}_{pi}^L\right) n_{\phi_{pi}^L} \\ &= \tilde{p}_i^L + \tilde{\phi}_{pi}^L n_{\lambda_{pi}^L} - \tilde{\lambda}_{pi}^L \left[\tilde{\phi}_{pi}^L\right]_\times N\left(\tilde{\phi}_{pi}^L\right) n_{\phi_{pi}^L} \\ &= \tilde{\lambda}_{pi}^L \tilde{\phi}_{pi}^L + J_\lambda n_{\lambda_{pi}^L} + J_\phi n_{\phi i}^L \end{aligned} \tag{43}$$

where

$$\begin{cases} J_\lambda = \tilde{\phi}_{pi}^L \\ J_\phi = -\tilde{\lambda}_{pi}^L \left[\tilde{\phi}_{pi}^L\right]_\times N\left(\tilde{\phi}_{pi}^L\right) \end{cases}$$

and $N(\tilde{\phi}_{pi}^L)$ is an orthonormal basis of the tangent plane at $\tilde{\phi}_{pi}^L$.

Based on the IMU forward propagation equation [23], the error state equation in (20) is written as

$$\begin{aligned} \delta_{\hat{x}_{I_{k+1}}} &= x_{I_{k+1}} \boxminus \hat{x}_{I_{k+1}} \\ &= \left(x_{I_k} \boxplus \left(\Delta t \mathrm{f}\left(x_{I_k}, \mathrm{u}_{I_k}, \mathrm{w}_{I_k}\right)\right)\right) \boxminus \left(\hat{x}_{I_k} \boxplus \left(\Delta t \mathrm{f}\left(\hat{x}_{I_k}, \mathrm{u}_{I_k}, 0\right)\right)\right) \\ &= F_{\delta \hat{x}_k} \delta \hat{x}_{I_k} + F_{\mathrm{w}_k} \mathrm{w}_{I_k}. \end{aligned} \tag{44}$$

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